

Baoli Hao

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🎓 [Baoli's Google Scholar](#)

EDUCATION

Illinois Institute of Technology

Ph.D. in Applied Mathematics

Aug. 2022 – Expected Apr. 2027

Chicago, IL

- **Advisors:** Prof. Chun Liu & Prof. Ming Zhong | **GPA:** 3.93/4.0

City University of Hong Kong

BSc in Computing Mathematics (First Class Honours)

Aug. 2017 – Apr. 2021

Hong Kong

- **GPA:** 3.7/4.3 | **Minor:** Computer Science

RESEARCH INTERESTS

My research focuses on **inverse problems** and **scientific machine learning (SciML)** for complex dynamical systems, with an emphasis on exploiting mathematical and physical structure for stable, interpretable, and scientifically meaningful data-driven learning and scientific computation. My work also connects these problems with **multiscale modeling**, **nonlinear PDEs**, and **interacting particle systems**.

PUBLICATIONS & PREPRINTS

1. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. A finite element framework for simulating residential burglary in realistic urban geometries. *Mathematical Models and Methods in Applied Sciences*, 36(5), 1019–1050, 2026.
2. Kin Shing Chan, **Baoli Hao**, Kei Fong Lam, and Björn Stinner. On a phase field model for binary mixtures of micropolar fluids with non-matched densities and moving contact lines. *Interfaces and Free Boundaries*, published online first, 2026.
3. **Baoli Hao**, Chun Liu, Ulisses Braga-Neto, Lifan Wang, and Ming Zhong. Stability in training PINNs for stiff PDEs: Why initial conditions matter. *Foundations of Data Science*, Early Access, 2026.
4. **Baoli Hao**, Ming Zhong, and Kevin O’Keeffe. Attractive and repulsive interactions in the one-dimensional swarmalator model. *Physical Review E*, 108(6), 064214, 2023.
5. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. Crime hotspot dynamics in residential burglary models with police response. arXiv:2605.17709, 2026.
6. **Baoli Hao**, Fanze Kong, Kei Fong Lam, and Chun Liu. Global finite-energy weak solutions and sharp entropy decay for a Poisson–Nernst–Planck system with interspecies drag and steric effects. arXiv:2607.21742, 2026.
7. **Baoli Hao**, Chenxi Hu, Ming Zhong, and Ren Wang. Event-structured physics-informed neural networks for differentiable critical clearing boundaries. arXiv:2607.27681, 2026.

MANUSCRIPTS

1. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. One shot of learning of collective dynamics from equilibrium in interacting particle systems. Manuscript in preparation, 2026.
2. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Learning interaction topology and kernels in interacting particle systems. Manuscript in preparation, 2026.
3. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Data-driven discovery of gravitational structure from solar-system ephemerides. Manuscript in preparation, 2026.

4. **Baoli Hao**, Annalisa Quaini, and Ming Zhong. Discovery of hidden crime dynamics, Part I: Parametric structures. Manuscript in preparation, 2026.
5. **Baoli Hao**, Xiaofan Li, Chun Liu, Yiwei Wang, and Ming Zhong. Data-driven discovery of local and nonlocal constitutive physics in ion transport. Manuscript in preparation, 2026.

TALKS & PRESENTATIONS

SIAM Annual Meeting	Jul. 2026
Title: <i>Modeling, Simulation and Learning for Urban Crime Dynamics</i>	Cleveland, OH
SIAM Texas-Louisiana Sectional Meeting	Sept. 2025
Title: <i>Stability in training PINNs for stiff PDEs: Why initial conditions matter</i>	Austin, TX
Central AMS Sectional Meeting	Apr. 2025
Title: <i>Modeling and analysis of Poisson-Nernst-Planck systems with steric effects and relative drags</i>	Lawrence, KS

INTERNSHIPS

HKU Business School, The University of Hong Kong	Jul. 2021 – May 2022
<i>Data Scientist</i>	Hong Kong
• Supervisor: Dr. Qingchen Wang • Developed machine-learning and predictive models for risk stratification and survival analysis using large-scale electronic patient record data covering approximately 90% of Hong Kongs population.	

SERVICE

<i>Coordinator</i> , <u>Multiscale Modeling and Computation Seminar</u> , Illinois Tech	Aug. 2022 – Present
<i>Sponsorship and Student Support Chair</i> , <u>ACM HyperText 2025 Conference</u> , Illinois Tech	Oct. 2024 – Sept. 2025
<i>Program Committee Member</i> , AAI Conference on Artificial Intelligence (AAAI 2027)	Aug. 2026
<i>Reviewer</i> , AAI Conference on Artificial Intelligence (AAAI 2026)	Sept. 2025
<i>Reviewer</i> , ACM Transactions on Modeling and Computer Simulation (TOMACS)	Jan. 2026

TEACHING

Illinois Institute of Technology	
MATH Graduate Qualifying Exam Analysis Prep Session (<i>Leader & Mentor</i>)	Fall 2025 – Present
MATH 500: Applied Analysis I (<i>Teaching Assistant</i>)	Fall 2025
MATH 578: Computational Mathematics II (<i>Grader</i>)	Spring 2025
MATH 151: Calculus I (<i>Recitation Instructor</i>)	Spring 2023

HONORS & AWARDS

Menger Lecture Poster Award , Illinois Institute of Technology	Apr. 2026
Full Tuition Scholarship , City University of Hong Kong	2017 – 2021
Deans List , City University of Hong Kong	2018 – 2021

SKILLS

Programming : Python, R, MATLAB, SQL
Frameworks & Libraries: PyTorch, TensorFlow, FEniCS, Scikit-learn, SciPy, NumPy, Chebfun
Languages : English (Fluent), Mandarin (Native), Cantonese (Good)
Others : LaTeX, Microsoft Office, Git